

KAIHUAI DENG

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EDUCATION

Nanjing University of Information Science and Technology (NUIST)

Nanjing, China

B.S. in Atmospheric Science (Hydrometeorology)

Sep 2023 - Jul 2027 (Expected)

NUIST ranked #1 in China and #8 globally for Atmospheric Science (2026 ShanghaiRanking GRAS).

- Weighted average: 85.53/100 | GPA: 3.57/4.0 | Class rank: 6/51 (**top 12%**)
- Key Laboratory of Hydrometeorological Disaster Mechanism and Warning of Ministry of Water Resources.
- Relevant coursework: Probability & Statistics, Remote Sensing for Hydrometeorology, Hydrometeorological Modeling, Numerical Weather Prediction, Python Programming.

RESEARCH INTERESTS

Remote Sensing · Hydroclimate Extremes · Land–Atmosphere Interactions · Climate Risk Analysis

PUBLICATIONS & MANUSCRIPTS

- [1] **Deng, K.**, Huang, Z.* *Area Expansion Dominates Rising Global Three-Dimensional Terrestrial Water Storage Drought Severity*. Under review at *Geophysical Research Letters*.
- [2] **Deng, K.**, Huang, Z.*, Li, H., Wu, J. *Global Unequal Exposure to Intensifying Hydrological Droughts*. Under review at *Journal of Hydrology* (R4 revision submitted).
- [3] Wang, J.*, **Deng, K.** (2024). *A Model that Explains the Contrasting SST Trends in the Southern Pacific Ocean. Science and Technology of Engineering, Chemistry and Environmental Protection*. <https://doi.org/10.61173/xnag2k18>

* Corresponding author.

CONFERENCE PRESENTATIONS

Deng, K., Huang, Z. (2026). *Area expansion dominates rising global three-dimensional terrestrial water storage drought severity*. AGU26, San Francisco, USA, 11 December 2026. Poster presentation; presenting author.

Global 3D TWS drought research. English research presentation, National Outstanding Undergraduate Forum, Department of Earth System Science, Tsinghua University, June 2026. (70 participants nationwide)

RESEARCH EXPERIENCE

Mitacs Globalink Research Internship

Charlottetown, PE, Canada

Research intern | Advisor: Prof. Xander Wang (University of Prince Edward Island)

Jul 2026 - Present

- Awarded the fully funded 2026 Chinese Scholarship Council (CSC)-Canada Mitacs Globalink Research Internship.
- Organized PEI Climate Hazard & Risk Information System shoreline and coastal protection data, prepared QGIS maps, and summarized shoreline change rates by coastal setting; co-authored an interim research report.

Global 3D Terrestrial Water Storage Drought Tracking

Sep 2025 - Present

Research assistant | Advisor: Prof. Zhongwei Huang (NUIST)

- Developed a three-dimensional event-tracking framework for 23 years of GRACE/GRACE-FO terrestrial water storage observations, identifying 672 drought events and reconstructing their splitting-merging lineages.
- Quantified area expansion, mean intensity, and interaction contributions to rising monthly drought severity (77.7%, 19.2%, 3.2%); compared event-scale precipitation and actual evapotranspiration anomalies.
- Principal Investigator of a Provincial Undergraduate Training Program on Innovation and Entrepreneurship.

Global Unequal Exposure to Intensifying Hydrological Droughts

Jun 2025 - Present

Research assistant | Advisor: Prof. Zhongwei Huang (NUIST)

- Applied EEMD to GRACE/GRACE-FO CSR RL06 mascon observations and derived event-based drought metrics; integrated GPW population grids to estimate a global trend of 38.76 million person-exposures yr⁻¹.

- Decomposed the exposure increase into hydroclimatic conditions (73.2%), population growth (5.3%), and their interaction.
- Led study design, global data processing, visualization, and manuscript drafting; R4 revision under review.

Fluctuation Theorem & TCWV Asymmetry

Sep 2024 - Sep 2025

Research assistant | Advisor: Prof. Jun Yin (NUIST)

- Extended a nonequilibrium fluctuation theorem (Yin et al., 2024, *J. Climate*) from surface temperature to TCWV using ERA5; quantified PDF asymmetry A_{asym} across $\tau = 1\text{--}20$ yr smoothing timescales.
- Validated SST–TCWV Clausius–Clapeyron scaling ($R^2 = 0.95$, approximately 7%/K) and tested the transferability of the tail-rescaling framework; produced a manuscript draft; discussed with Prof. Lamberto Rondoni (Politecnico di Torino).

WRF-based Precipitation Sensitivity over Jing-Jin-Ji

Mar 2025 - Jul 2025

Advisor: Dr. Mengya Wang (NUIST)

- Configured WRF/WPS with three nested domains on Linux HPC; conducted 16 experiments perturbing soil moisture ($\pm 10\%$ to $\pm 50\%$) and urban LULC to represent rapid-urbanization scenarios.
- Evaluated precipitation responses using spatial correlation, RMSE, and threat scores; identified the lowest-bias cumulus–microphysics configuration among the tested experiments.

Precipitation–Runoff Prediction in the Yiluo River Basin

Jun 2024 - Mar 2025

Research assistant | Advisor: Prof. Xing Yuan (Institute of Atmospheric Physics, Chinese Academy of Sciences)

- Applied EOF to monthly precipitation interannual increments, extracting three PCs explaining >90% of variance; screened hydroclimatic predictors using correlation analysis and stepwise regression.
- Built a BiLSTM + interannual-increment hybrid model, improving TCC by 16% and reducing RMSE by 20% relative to statistical downscaling; evaluated CSSPv2 monthly streamflow forecasts.

Contrasting SST Trends in the Southern Pacific Ocean

Mar 2024 - Sep 2024

Student researcher | Advisor: Prof. George Tselioudis (Columbia University / NASA GISS)

- Analyzed 29 years of multi-source satellite and reanalysis data (Copernicus SST, DUACS SSH, CERES, NCEP, NSIDC SIC) across the Southern Pacific (80°W–180°, 70°S–30°S).
- Derived Ekman transport (U_E, V_E, W_E) from wind stress to attribute the SST dipole and diagnosed a sea-ice radiative positive-feedback cycle; published as co-author in *STECEP* (2024).

INTERNSHIP

Leizhou Meteorological Bureau, Guangdong

Jul 2025 - Aug 2025

Intern | Forecast Division

Zhanjiang, China

- Supported operational monitoring and warning for Typhoon Danas (2025) and Typhoon Wipha (2025), tracking intensification and landfall trajectories and comparing ECMWF, CMA, NCEP, and TRAMS forecasts.
- Participated in calibration and telemetry checks for meteorological field instruments.

SELECTED AWARDS & HONORS

Provincial Undergraduate Training Program: Principal Investigator (provincial-level project) May 2026

Mitacs Globalink Research Internship Award: CSC-Mitacs Joint Program (≤ 260 awardees nationwide) Jan 2026

Second Class Academic Scholarship: NUIST (top 20%) Nov 2024 & Nov 2025

Merit Student: NUIST (top 15%) Dec 2024

ADDITIONAL INFORMATION

Programming: Python, Shell scripting, Fortran, Typst/LaTeX

Scientific Computing: Google Earth Engine, Linux/HPC, Git, WRF/WPS, MATLAB, ArcGIS, QGIS

Machine Learning: TensorFlow, PyTorch, BiLSTM, EOF, EEMD

Languages: Mandarin (native), Cantonese (native), English (proficient), French (basic)

Academic Service: Creator and maintainer of [othodeng.github.io](https://github.com/othodeng), publishing study notes from 10+ courses